

Basic Principles of Medicine 1

Module: Cardiovascular, Pulmonary, Renal | DLA 18

Clinical Anatomy of the Urinary System – Upper Urinary Tract

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Drake, Vogl, Mitchell. Gray's Anatomy for Students.
5th edition. Elsevier Churchill Livingstone.

- Online Textbook Chapter 4 (Abdomen) & Chapter 5 (Pelvis and Perineum):
 - Conceptual Overview
 - Regional Anatomy

***Pay special attention to the “In the Clinic” boxes
for the clinical correlations for this topic***

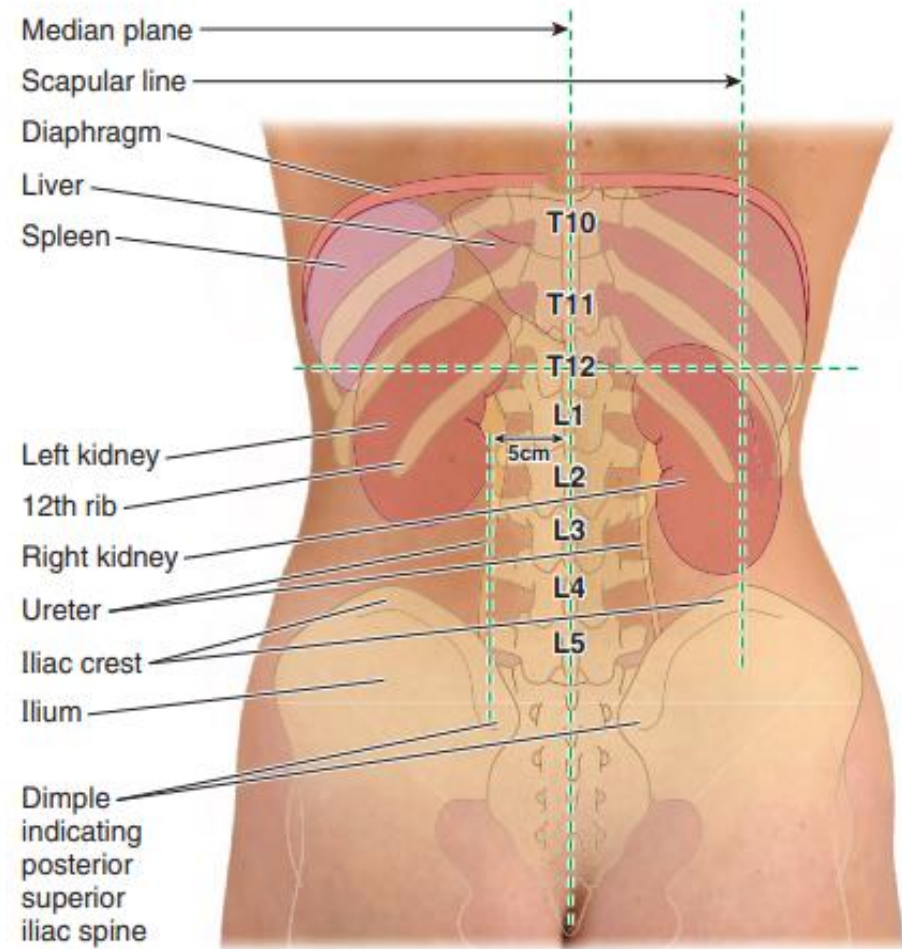


Objectives

Describe the location of the kidneys and their relationship with surrounding organs and ribs 11 and 12	SOM.MK.II.BPM1.3.CPR.2.ANAT.1358
Describe the relationship of the kidneys to the diaphragm, psoas major, quadratus lumborum and transversalis abdominis aponeurosis	SOM.MK.II.BPM1.3.CPR.2.ANAT.1359
Describe the anatomy of the cortex and medulla of the kidney	SOM.MK.II.BPM1.3.CPR.2.ANAT.1360
Identify the structures in the hilum of the kidney and their relationship to one another	SOM.MK.II.BPM1.3.CPR.2.ANAT.1361
Describe the blood supply, innervation and venous drainage of the kidney	SOM.MK.II.BPM1.3.CPR.2.ANAT.1362
Identify the course, relationships, blood supply and innervation of the ureters	SOM.MK.II.BPM1.3.CPR.2.ANAT.1472
Describe the kidneys in relation to the renal fascia, perirenal and pararenal fat	SOM.MK.II.BPM1.3.CPR.2.ANAT.1473
Describe the course of the ureter along the posterior abdominal wall to the bladder	SOM.MK.II.BPM1.3.CPR.2.ANAT.1475
Describe the structures which support the urinary bladder	SOM.MK.II.BPM1.3.CPR.2.ANAT.1477
Describe the internal anatomy of the urinary bladder	SOM.MK.II.BPM1.3.CPR.2.ANAT.1478
Describe the blood supply, venous drainage, lymphatic drainage and innervation of the urinary bladder	SOM.MK.II.BPM1.3.CPR.2.ANAT.1479

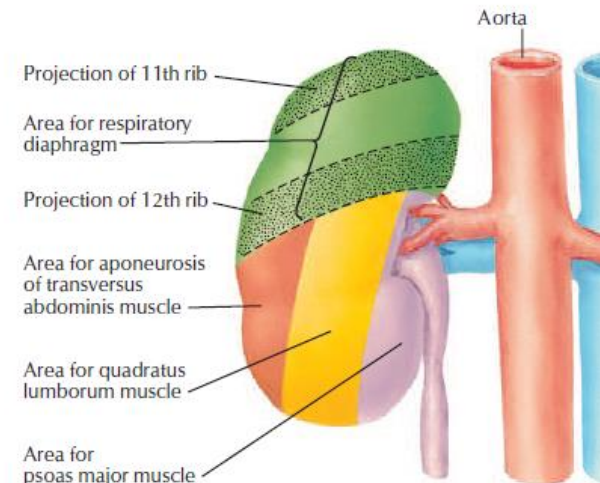
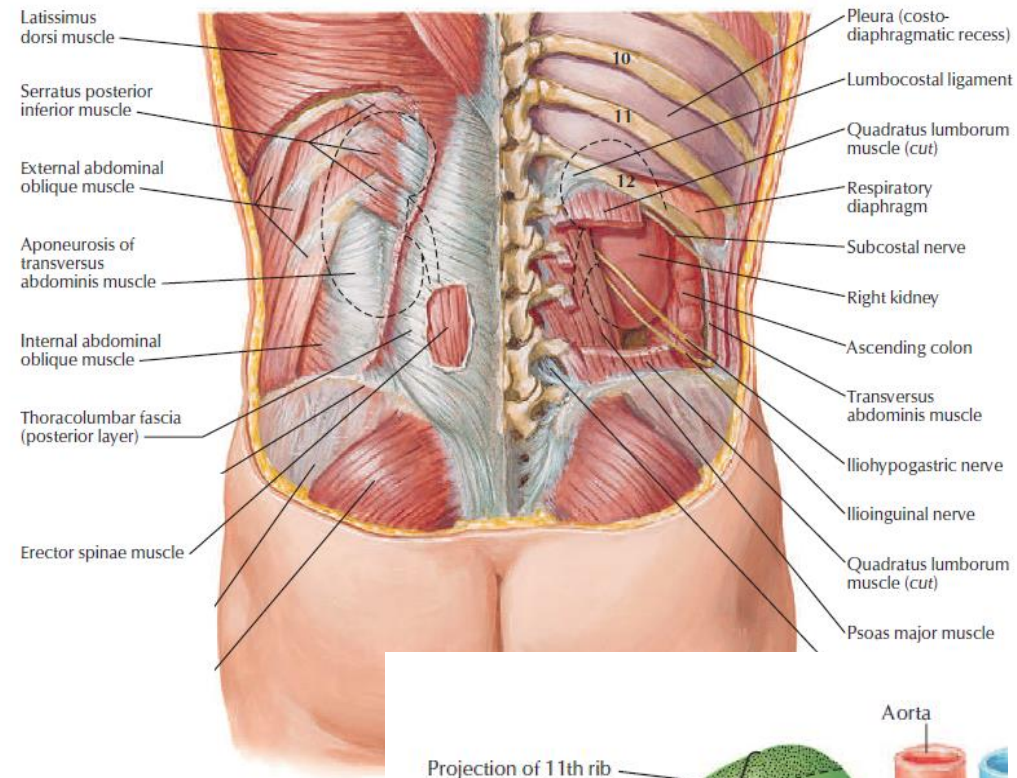
Kidney - Surface Reference Points

- The kidneys are located at the level of T12-L3 vertebrae
- Transpyloric plane:** imaginary horizontal line around L1
 - Right kidney lies just below the plane
 - Left kidney lies just above the plane
- Renal hila located about **5 cm** from midline
 - Hilum:** a concavity located on the medial border of each kidney where structures enter and leave
 - Transmits **renal artery, renal vein, nerves** and **renal pelvis**



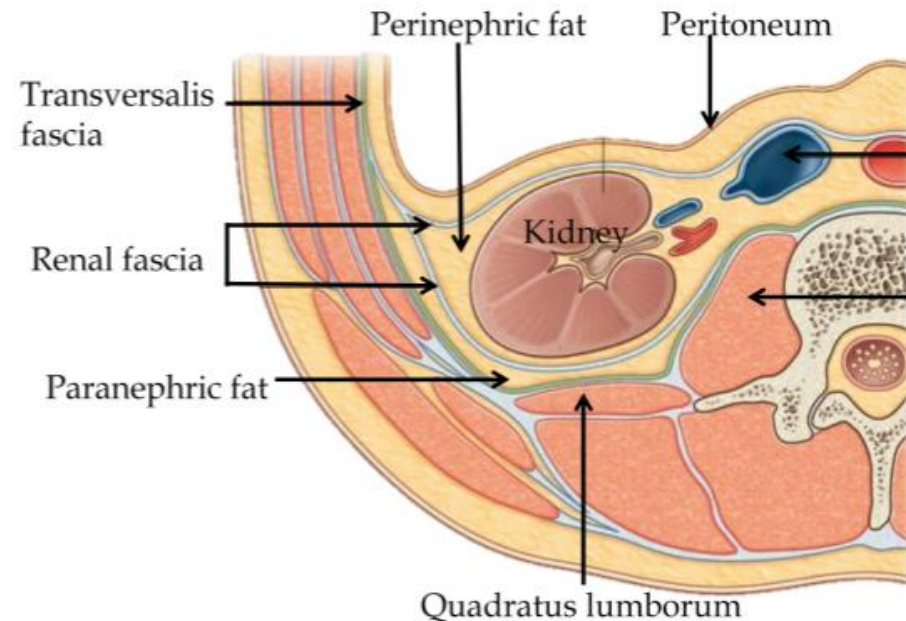
Kidney – Superficial Anatomy Associations

- Kidneys lie mainly under the cover of the costal margin
 - accessible between the costal margin and vertebral column (i.e. **costovertebral angle**)
- Superiorly associated with diaphragm
 - separates kidneys from the pleural cavities and 11th/12th ribs
- Inferoposterior surfaces are related to the muscles of the posterior abdomen
 - **quadratus lumborum muscle** as well as **psoas major** and the aponeurotic tendon of **transversalis abdominis**



Kidney – Enveloping Structures

- Renal capsule
 - Connective tissue closely adhered to the external surface of the kidney
- Perirenal (perinephric) fat
 - Layer of fat enclosing the renal capsule
- Renal fascia (of Gerota)
 - Membranous condensation of extraperitoneal fascia
 - Encloses perirenal fat and the adrenal gland
- Pararenal (paranephric) fat
 - Layer of fat outside of the renal fascia

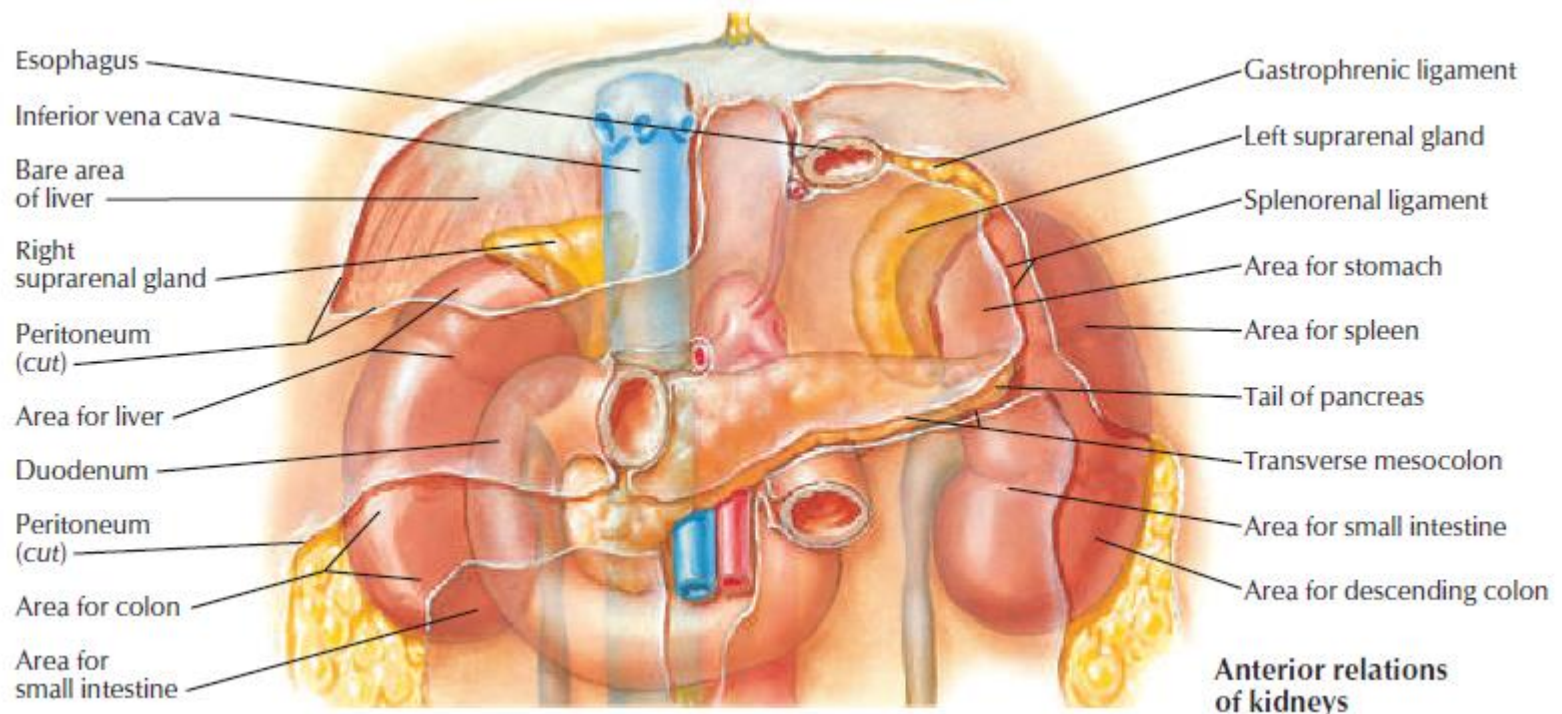


Kidneys In Situ

- Kidneys lie high up on the posterior abdominal wall
 - Located behind the peritoneum (i.e. **retroperitoneal**)

Anatomical relations

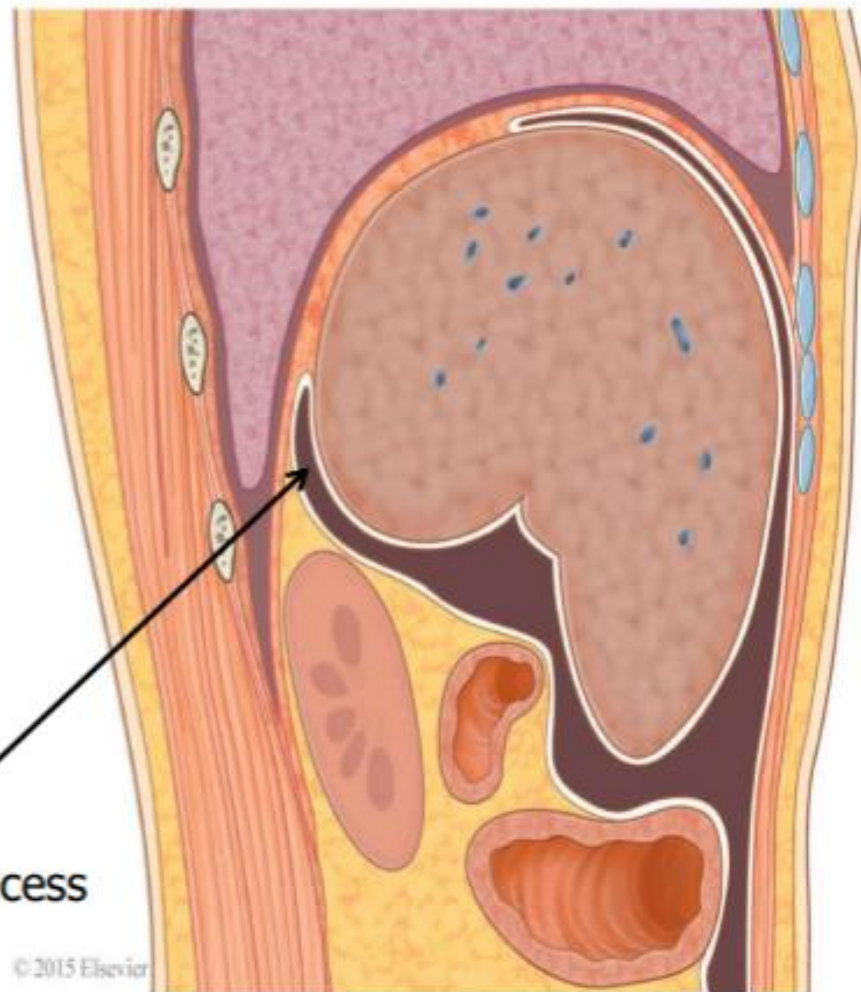
- Superiorly: adrenal glands, liver (on right), spleen (on left)
- Medially: duodenum (on right), tail of pancreas (on left)
- Inferiorly: right or left colic flexure, jejunum (on left)



Hepatorenal Recess

- AKA "pouch of Morrison"
- Located superior to the right kidney and posterior to the liver
- Fluid in the abdomen will accumulate in this space when the body is supine
 - requires investigation following trauma
 - viewed as part of the *focused assessment with sonography in trauma (FAST)* examination

Hepatorenal recess

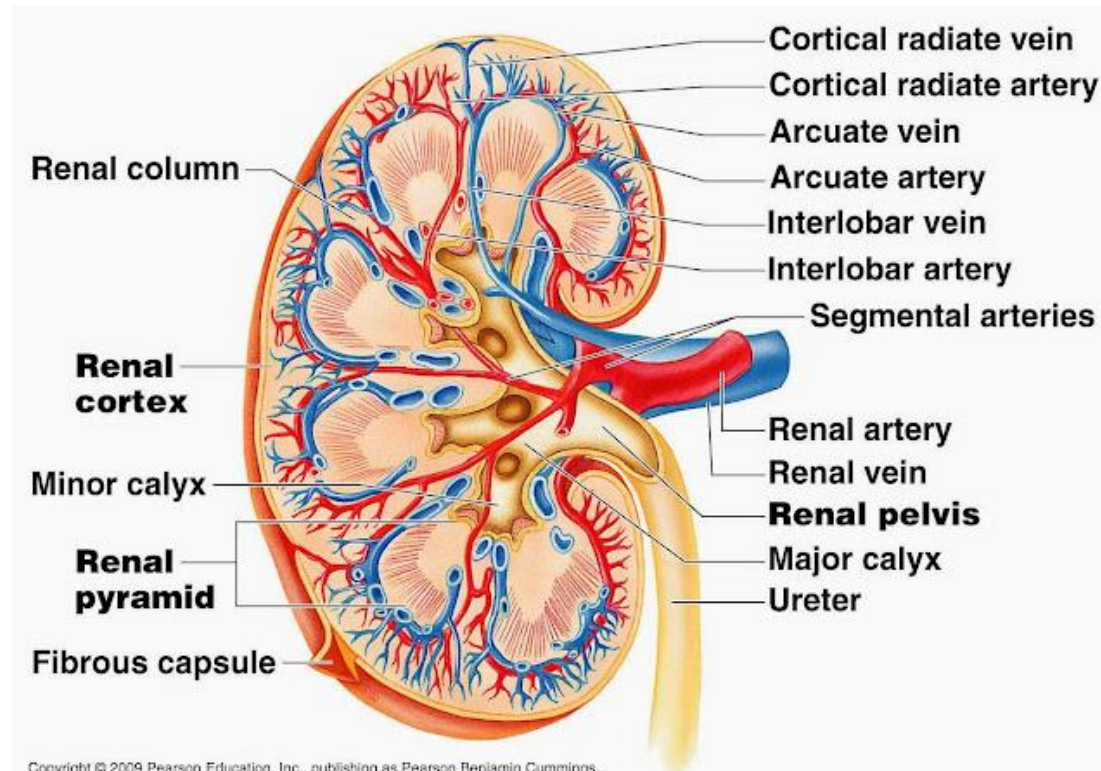


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Renal Capsule

- Protective layer surrounding the outer kidney
 - Composed of collagen, elastic and smooth muscle fibers
- Penetrates the renal parenchyma to form trabeculae
 - Lobulates the kidneys
- Enters the renal hilum and lines the renal sinus
 - Continuous with the connective tissue forming the walls of the calyces and renal pelvis



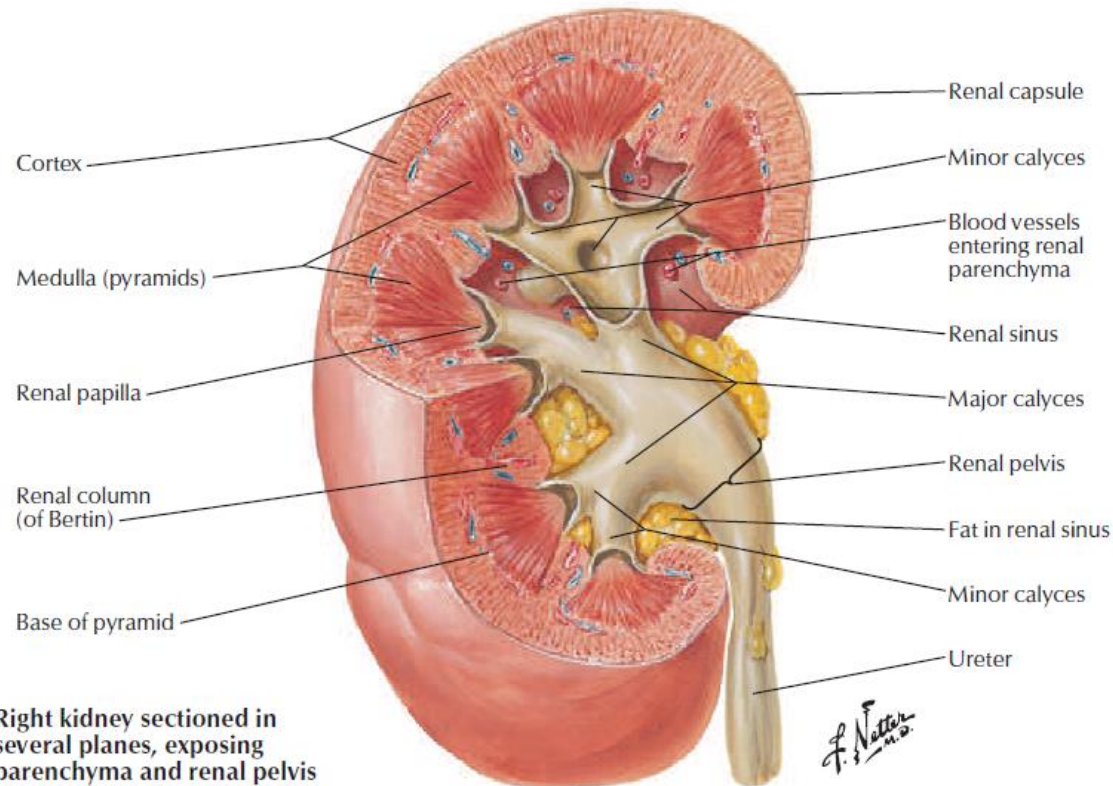
Renal Parenchyma

Renal cortex

- subcapsular region that arches over bases of renal pyramids
- extends as renal columns (of Bertini) between the renal pyramids

Renal medulla

- layer composed of 8-16 pyramids
- **Papilla**: apex of a renal pyramid
 - projects into the renal sinus
 - surface is perforated by the collecting ducts (of Bellini)



Renal Blood Supply

• Renal artery

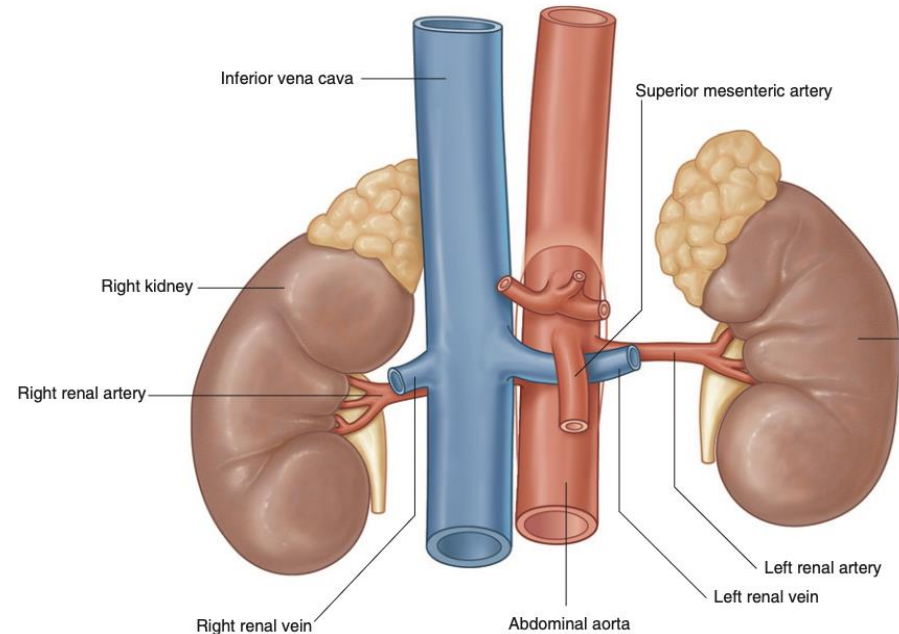
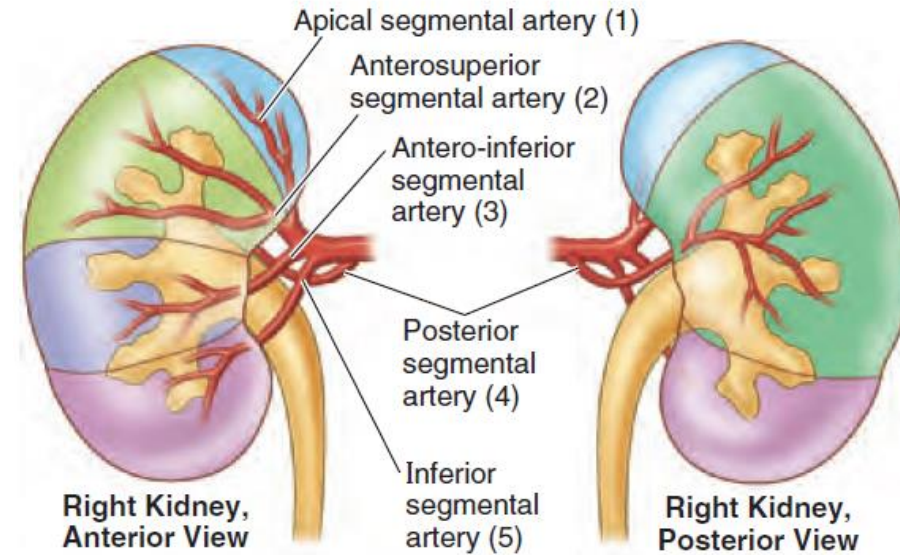
- arises at the level of the IV disc between L1 and L2 vertebrae
- Typically, each artery divides close to the hilum into five segmental arteries, which are end arteries.

• Renal vein

- Drains the kidney and subsequently drains into the inferior vena cava
- **Right renal vein** (2.5 cm) receives tributaries exclusively from the kidney
- **Left renal vein** (7.5 cm) receives left suprarenal vein and left gonadal (testicular or ovarian) vein

• Anatomical relationships

- renal veins lie anterior to the renal arteries
- Longer **right renal artery** passes posterior to IVC
- Longer **left renal vein** traverses the acute angle between the *superior mesenteric artery* anteriorly and *aorta* posteriorly



Ureters

- 25 cm long muscular ducts that carry urine from the kidneys to the urinary bladder
- Begin at the **ureteropelvic junction**, posterior to the renal vein and artery in the hilum

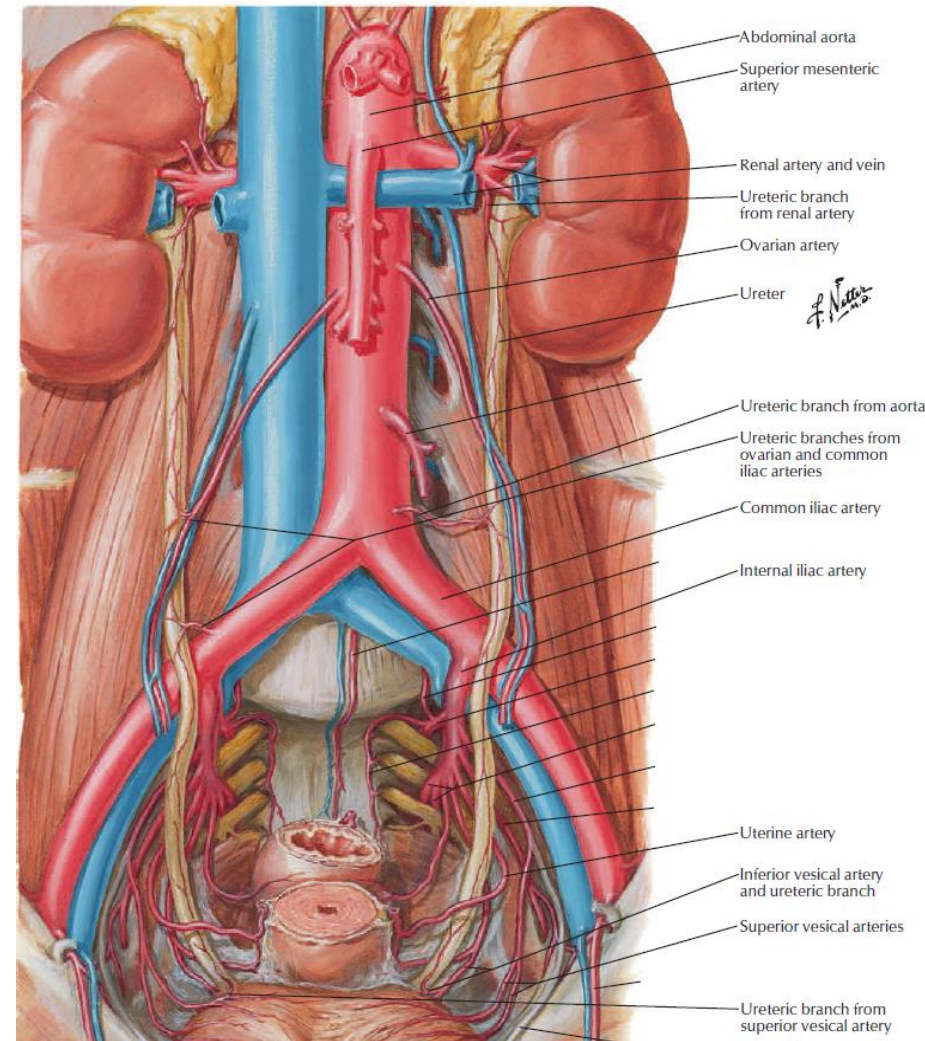
Route from kidneys to urinary bladder consists of three parts

- Proximal (abdominal):
 - travels retroperitoneally from renal pelvis to pelvic brim
- Pelvic:
 - from pelvic brim to the urinary bladder
- Distal (intramural):
 - enters urinary bladder wall and ends at ureteric orifices



Ureter Course

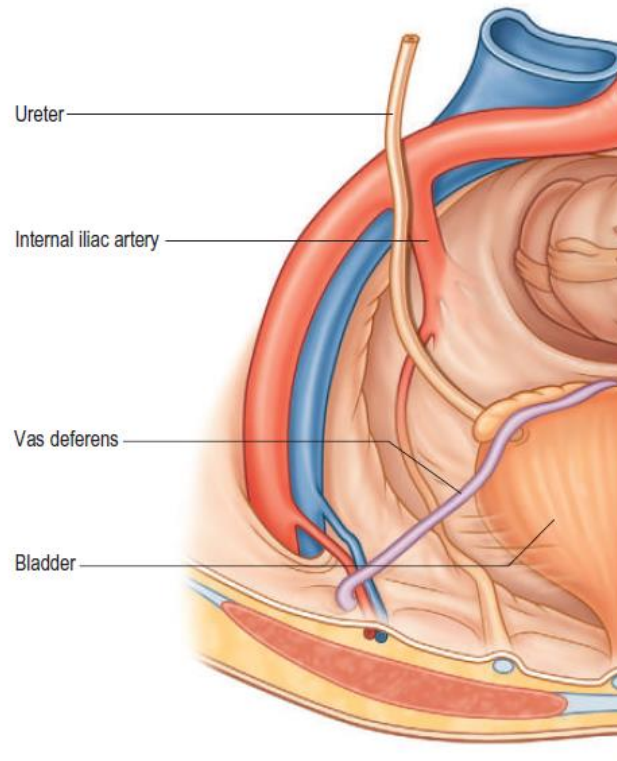
- Proximal (abdominal) part:
 - descends over psoas major
 - crosses anterior to the genitofemoral nerve
 - obliquely crossed by the gonadal vessels
- Pelvic part:
 - enters anterior to the end of common iliac vessels
 - different pelvic course for males and females
- Distal (intramural) part:
 - runs obliquely through the urinary bladder wall



Ureter Course within the Pelvis

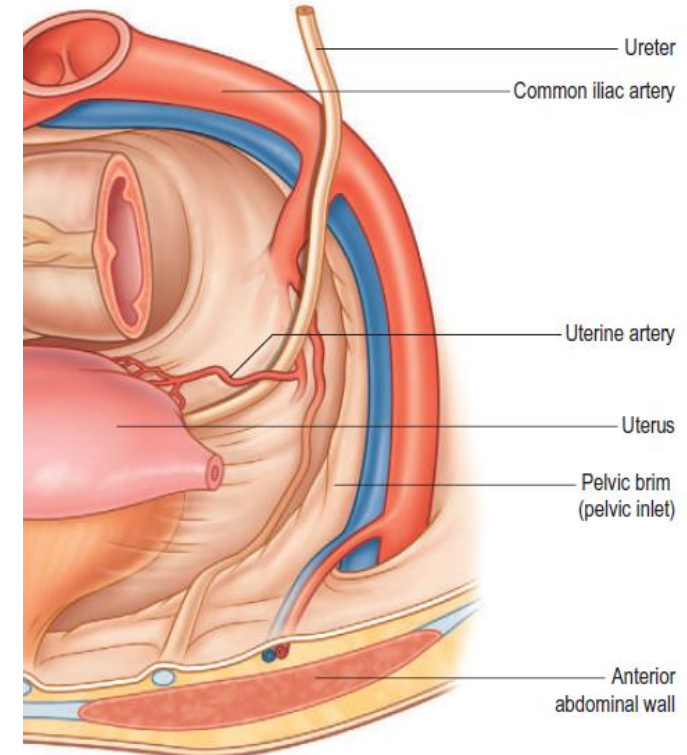
Males:

- Ureter travels under ductus deferens
- Then ureter passes anterosuperiorly to upper pole of seminal gland



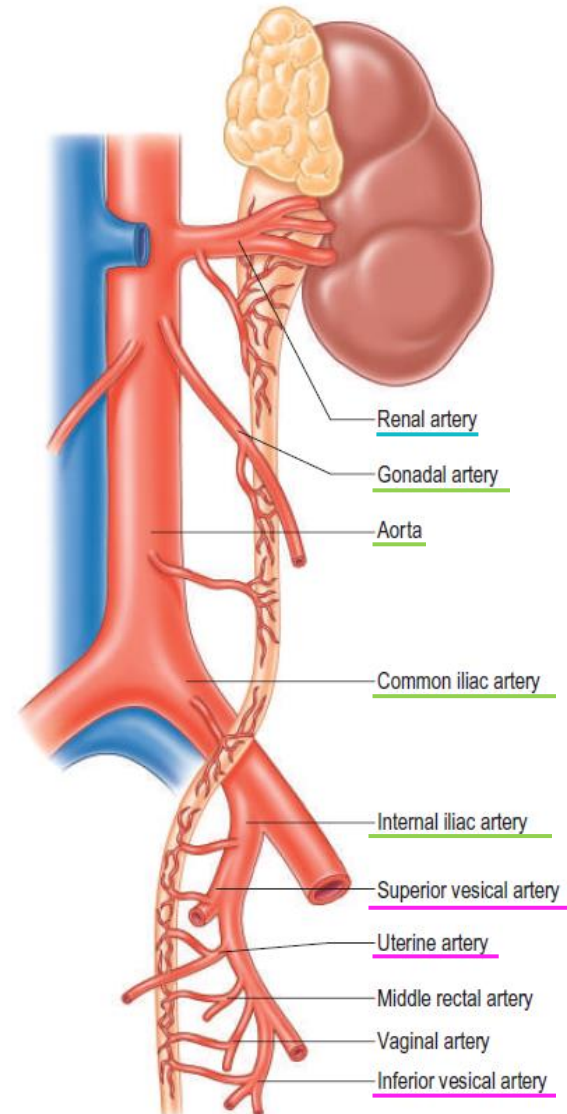
Females:

- Uterine artery is initially anterosuperior to the ureter
- Then ureter crosses medially and ascends the uterus before reaching the bladder



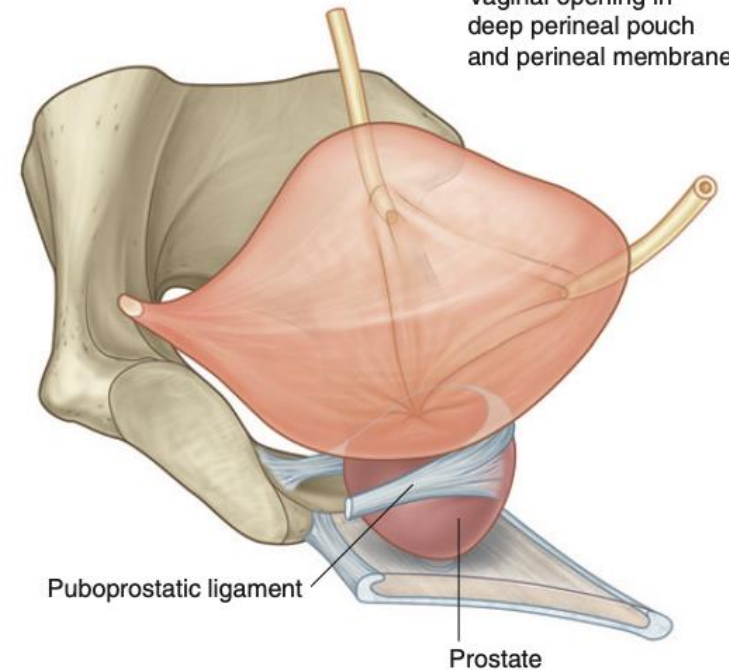
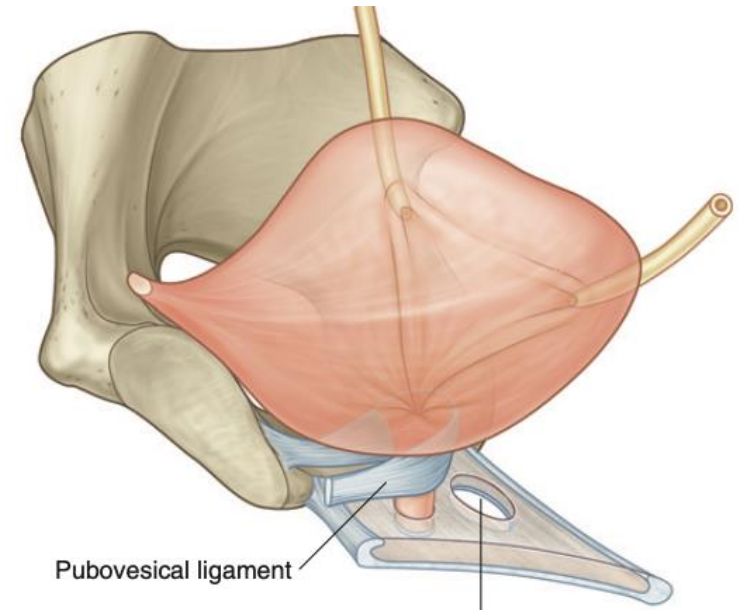
Blood Supply of Ureters

- Upper end supplied by
 - ureteric branch of the renal artery
- Middle portion supplied by branches from
 - abdominal aorta
 - gonadal artery
 - common iliac and internal iliac arteries
- Lower end supplied by branches from
 - inferior and superior vesical artery
 - uterine artery



Urinary Bladder

- A hollow viscus with strong muscular walls, that is characterized by its distensibility
- When empty, the adult urinary bladder is in the *lesser pelvis*
 - partially superior and posterior to the pubic bones
- The bladder is relatively free within the extraperitoneal subcutaneous fatty tissue
- However, the bladder neck is held firmly by
 - **lateral ligaments of bladder**
 - **tendinous arch of the pelvic fascia**
 - pubovesical ligament in females
 - puboprostatic ligament in males



Urinary Bladder Anatomy

Apex

- located superiorly, directed towards the pubic symphysis
- connected to the umbilicus by the **median umbilical ligament**

Fundus (Base)

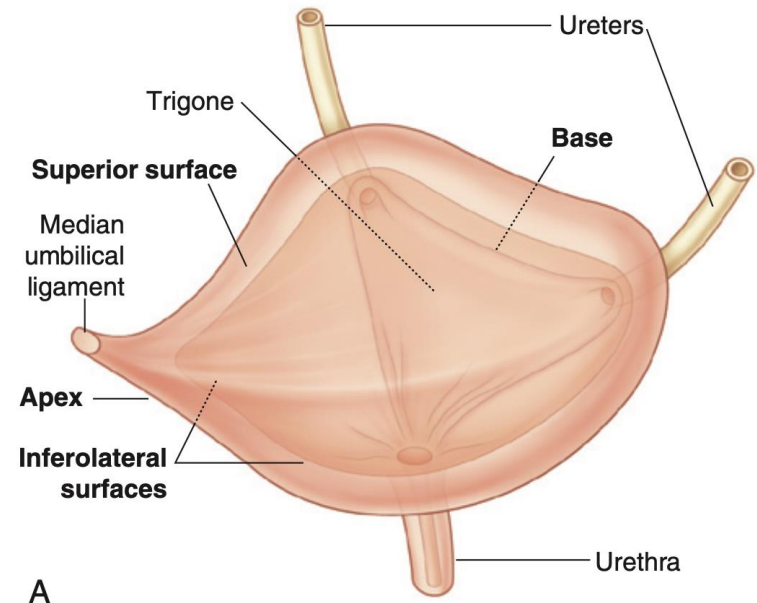
- inverted triangle that faces posteroinferiorly
- two ureters enter at upper corners
- **urethra** drains from lower corner
- **Trigone**: smooth area between these openings
 - Bulk of trigone is derived from **detrusor** - bladder smooth muscle

Body

- main part of the bladder
- located between the apex and the fundus

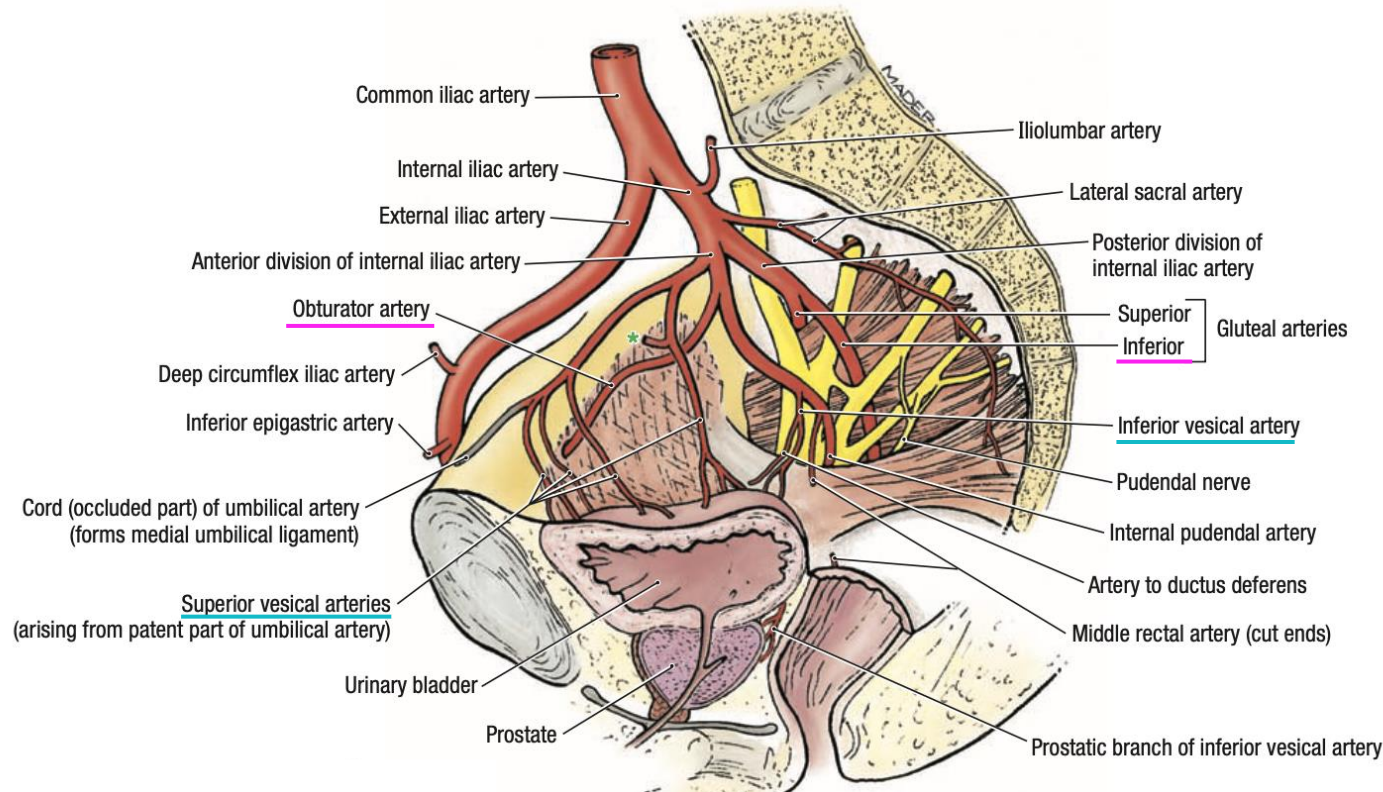
Neck

- formed by the convergence of the fundus and the inferolateral surfaces
- continuous with the **urethra**, which excretes urine outside of the body



Urinary Bladder - Blood Supply

- The **superior and inferior vesical arteries** provide most of the arterial blood but there are small contributions to the lower part of the bladder from the **obturator, inferior gluteal**, as well as **uterine and vaginal arteries**.
- The veins draining blood from the bladder correspond to the arteries and are tributaries of the **internal iliac veins**.
 - Lymphatics that drain the bladder mostly end in the **external iliac nodes**



Innervation

- Sympathetics to kidney, ureters, and bladder are derived from thoracic splanchnic and lumbar splanchnic nerves
- Afferent fibers, including those subserving pain, accompany sympathetic nerves as they do for most other viscera.
 - Some afferents may run with **vagal fibers**.

Kidneys: preganglionics at T12-L1 levels

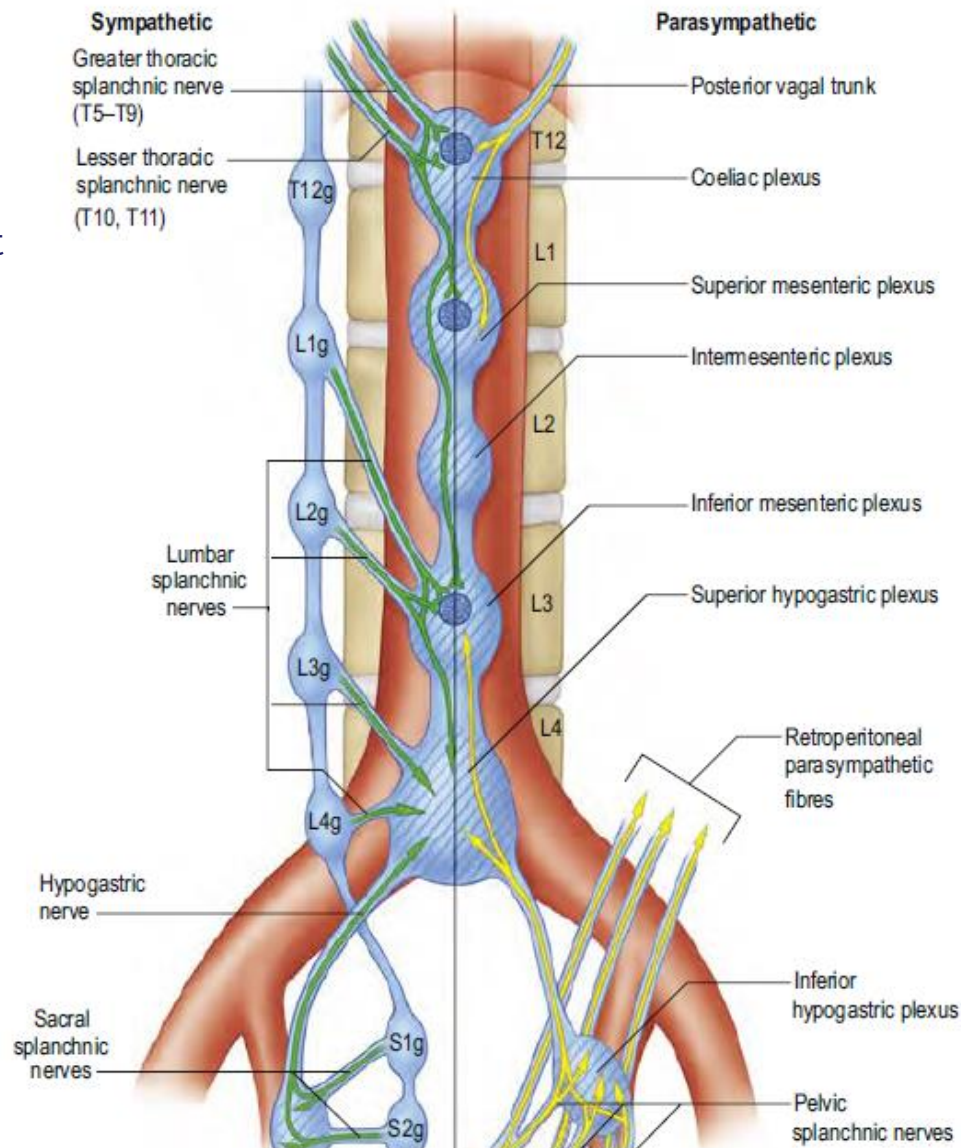
- celiac plexus**
 - greater and lesser thoracic splanchnics
 - vagal trunks**
- first lumbar splanchnic nerve

Ureters: T11-L2 spinal cord levels

- segmental innervation through nerves that follow blood vessels
 - Upper part:* **celiac plexus**
 - Middle part:* superior hypogastric plexus
 - Lower part:* **inferior hypogastric plexus**

Bladder

- includes **inferior hypogastric plexus**



Lymphatic Drainage

Kidney

- Left and right lumbar lymph nodes

Upper ureter

- Kidney lymphatics or directly to lumbar lymph nodes

Middle ureter

- Common iliac lymph nodes

Lower ureter

- Common, external, and internal iliac lymph nodes

Pelvic ureter and bladder

- Internal iliac lymph nodes

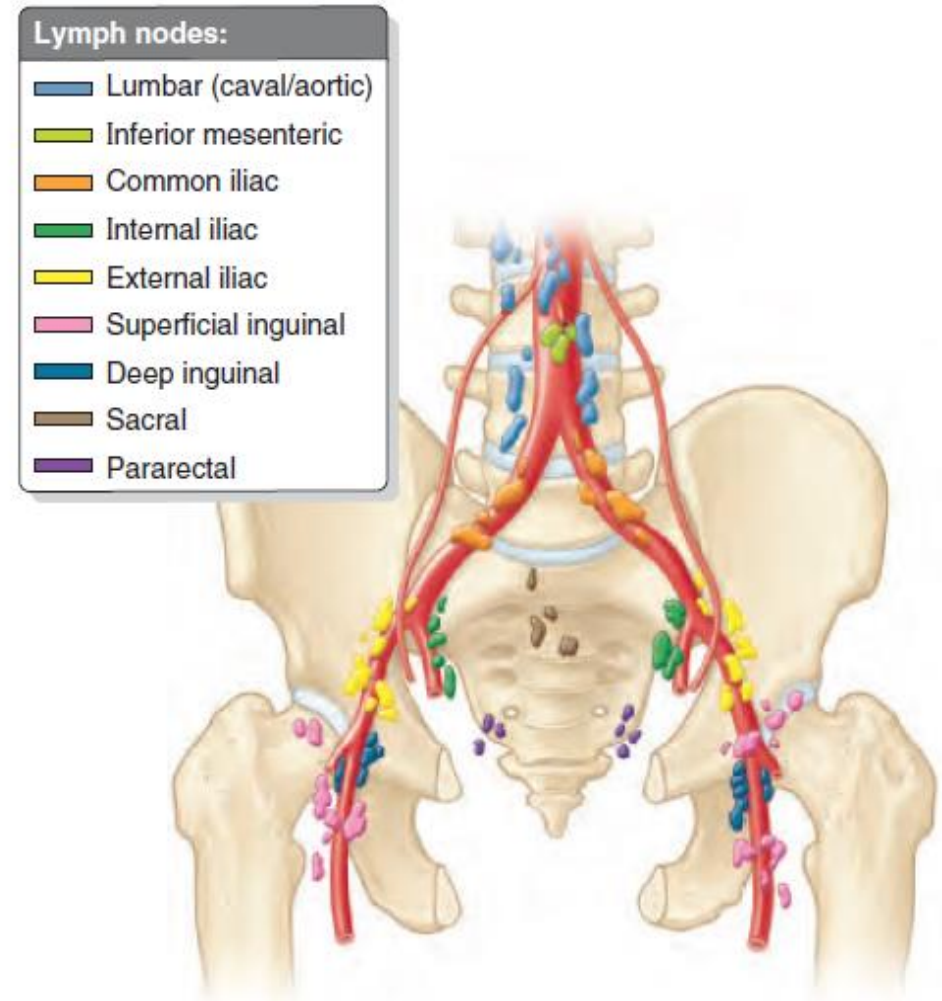


FIGURE 3.20. Lymph nodes of pelvis.



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